

JOSE DAVID CHAY GRIJALVA

Full-Stack Developer

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[GitHub](#) | [Linkedin](#) | [Portfolio](#)

EDUCATION

Universidad San Carlos de Guatemala

Mayor Systems Engineering

Guatemala City

January 2021 - Present

Instituto Guillermo Putzeys Alvarez

High School Degree

Guatemala City

January 2019 - October 2020

EXPERIENCE

Telus Digital Solutions | Full-stack Engineer

Guatemala City - Remote | Nov 2024 - Present

Contributed as a full-stack developer for a high-profile U.S. client (confidential). Focusing on the development of data-centric web applications using React, Firebase, Firestore, and Firebase Functions. I have implemented advanced state management solutions using Redux and Zustand, and developed comprehensive form-based workflows with Jest for testing and Playwright for QA automation. Additionally, I designed and configured automated CI/CD pipelines using GitHub Actions. My responsibilities extend beyond frontend development, involving Python scripting for data processing, including image and video manipulation, as well as the generation of analytical reports. This role has strengthened my experience with cloud infrastructure, particularly Google Cloud Platform services such as IAM, Cloud Functions, and Cloud Storage, while also leveraging AWS S3 for efficient long-term storage solutions. My work directly supports the client's data collection and analysis initiatives, contributing to critical decision-making processes.

MultiGroup | Frontend Engineer

Guatemala City | March 2024 - Nov 2024

Specialized in building and optimizing web and mobile interfaces using TypeScript, React.js, Next.js and React Native.

Additionally, I've provided support and improved the performance of Django applications by optimizing queries, consuming store procedures and programing async events with Celery, achieving a 75.08% improvement in loading speed.

Successfully implemented state management for web applications using Zustand and Redux, utilizing Reselect to optimize memoization, resulting in improved performance and reduced unnecessary re-renders.

Worked on a document digitalization project, transitioning physical SAP documents to a web application. The project involved integrating a PowerShell script, Node.js script, Express.js backend with MongoDB, and a frontend built with Next.js.

Grupo PB | Web Developer

Guatemala City | Jul 2022 - March 2024

Developed and maintained web applications using React.js and Next.js, ensuring high performance and responsiveness.

Led the integration of AWS services, including S3 and CloudFront, to optimize content delivery for frontend applications. Collaborated closely with backend developers, enhancing API interactions and improving data flow within the frontend.

Fiverr | Freelance Programmer

Remote | Feb 2022 - Jun 2022

Developed responsive websites using HTML, CSS, and JavaScript, focusing on creating clean, user-friendly designs.

Implemented custom JavaScript solutions for various client needs, improving functionality and interactivity. Worked on full-stack projects, primarily focusing on the frontend, ensuring seamless integration with backend services.

SKILLS

Programming Languages: JavaScript, TypeScript, Python, SQL, HTML/CSS

Libraries/Frameworks: React.js, Next.js, Angular, Express.js, Nest.js, Django, Selenium, Playwright, Beatiful soup

Tools / Platforms: AWS, Azure, Google Cloude, Git/GitHub, Docker

Databases:

MySQL, PostgreSQL, MongoDB, DynamoDB, Firebase, Firestore

PROJECTS / OPEN-SOURCE

Recognition IA | [Link](#)

Python, FastAPI, PostgreSQL, AWS (EC2, S3, RDS), Next.js,

Recognition IA is a full-stack facial recognition system designed as a freelance project, built entirely in Python and deployed using modern cloud infrastructure. The architecture consists of three core backend services and a client-facing web interface:

+ *IP Camera Capture*: A long-running Python script that captures real-time video from surveillance cameras. It extracts image frames every second and sends them via WebSocket to the recognition backend.

+ *Recognition_API*: A high-performance FastAPI backend designed with Hexagonal and Screaming Architecture. This service processes incoming images using DeepFace and RetinaFace to detect gender, estimated age, emotion probabilities (as a dictionary of confidence scores), and facial embeddings. The processed data is sent to the database service via REST API, while the raw images are uploaded to an AWS S3 bucket with automated cleanup (images are deleted after 48 hours).

+ *Recognition_DB*: A RESTful API built with FastAPI and PostgreSQL to store and retrieve recognition data. It supports basic CRUD operations (primarily GET and POST).

Additionally, I developed a Next.js web interface styled with Chakra UI, simulating a Google Meet-like experience. This frontend captures webcam input and sends frames via WebSocket to the *Recognition_API*, which responds with real-time demographic and emotional data to be displayed in the UI.

The system was deployed using AWS services: EC2 (with compute-optimized instances for recognition), RDS for PostgreSQL, S3 for image storage, and ECR for container management. GitHub Actions and AWS CodeDeploy were used for CI/CD automation. In some cases, deployments were also carried out manually to client-owned Linux servers.

RRHH System | [Link](#)

>Frontend: React.js, Sass > Backend: Python

Developed a complete HR management system with modules for Employees, Departments, and Job Positions. The frontend was built using Angular, providing a clean and responsive user interface for managing employee records, departments, and job positions. The system allowed for CRUD operations across all modules, utilizing dynamic forms and data fetching with RESTful API integrations.

The backend was developed using Python and Flask, structured with modular blueprints, and implemented best practices for API design, including validation, exception handling, and interaction with a PostgreSQL database.

BACKEND LINK: https://github.com/Josepe20/rrhhsystemtest_api

CERTIFICATIONS

- AWS Cloud Practitioner - **Amazon Web Services**
- 100 Days of Code: The Complete Python Pro Bootcamp for 2023 - **Udemy, Angela Yu**
- TypeScript Crash Course - **Udemy, Jan Schaffranek**
- Curso Bsico de Django - **Platzi, Facundo Garcia Martoni**
- Curso Prctico de SQL y MySQL - **Platzi, Alberto Alcocer**
- Curso Profesional de Git y GitHub - **Platzi, John Freddy Vega**
- Curso Profesional de Scrum - **Platzi, Gerardo Romero**
- English B2, Working With Clients - **British Council**
- Gestion de datos en Excel - **Camara de comercio y capacitacion international**